

Appendix

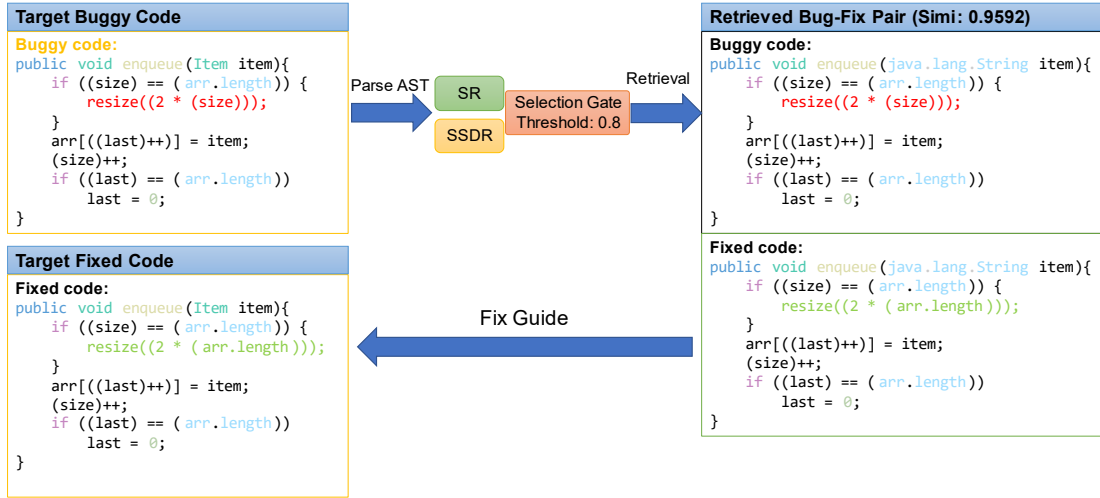


Figure I: A Real Example of How SelRepair Works

We present a real buggy code snippet in the test set and show how SelRepair fixes this buggy code. The **size** variable in line 3 needs to be replaced with variable **arr.length** to ensure the rationalization of array expansion. The detailed fixing process is shown in Figure I. When SelRepair receives the buggy code, the code will be parsed into AST. The code and AST will be input to the SR and SDR module to get the feature vector and calculate the similarity with each sample in the codebase. Then we adopt the selection gate and set the similarity threshold to 0.8. A bug-fix pair in the codebase is retrieved and the similarity is 0.9592. The bug-fix pair provides a similar fix pattern so SelRepair can fix the bug with the use of this RAG information.